

James Friedenber

Full-Stack Engineer & Platform Engineer

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[LinkedIn](#) | [Portfolio](#) | [Github](#)

SUMMARY

High-impact Software Engineer with 7+ years of experience building scalable live-event systems, cloud infrastructure, and developer tooling at Microsoft. Proven track record of driving **8x server capacity increases**, **cutting Azure infrastructure costs by 88%**, and delivering automated pipelines that save hundreds of engineering labor hours per release cycle. Expert in full-stack TypeScript/React ecosystems and high-scale Azure cloud environments.

PROFESSIONAL EXPERIENCE

Software Engineer, Microsoft - Remote, USA

March 2022 - June 2026

- Engineered Minecraft: Bedrock Edition's first automated multiplayer performance testing framework, **achieving an 8x increase in server capacity** (10 to 80 players) and **cutting Azure infrastructure costs by 88%** (~\$20k+ saved per live event).
- Developed the foundational prototype and gameplay systems for the Minecraft Legends promotional live event, **reaching 3M+ players and driving an estimated \$6M in marketing value** for the franchise launch.
- Modernized the MCTools codebase and expanded creator content validation suites, **replacing manual workflows with automated pipelines to cut hundreds of engineering labor hours per release cycle**.
- Designed and implemented a telemetry and observability framework for MCTools using Azure Application Insights, enabling custom event instrumentation, real-time monitoring, issue detection, and operational visibility across the platform.
- Developed a localization framework, enabling MCTools to be available in 20+ languages and bringing it into compliance with the minecraft.net domain.
- Mentored and onboarded new engineers to the Minecraft web and platform technology stack, providing guidance on React, TypeScript, deployment workflows, telemetry tooling, and development practices.

Software Design Engineer in Test, DIGARC - Remote, USA

March 2021 - March 2022

- Developed detailed test plans and tracked quality initiatives using Jira for an Agile team of developers, testers, and a PM.
- Created and maintained several hundred automated E2E tests, ensuring reliable frontend regression coverage.
- Led weekly team meetings to evaluate performance and functionality metrics in preparation for production deployments.

Software Design Engineer in Test, Andromeda Systems Inc. - Jacksonville, Florida

August 2019 - March 2021

- Developed over 500 automated tests using Selenium for an inventory management application used by the U.S. Navy for aircraft maintenance.
- Built dozens of Angular components using TypeScript for administrative tools to improve client operations and workflows.
- Ensured product quality in an Agile environment by triaging bugs and writing user stories and test cases.

QA Automation Engineer CO-OP, LenelS2 - Rochester, New York

January 2018 - August 2018

- Developed and designed a cloud-based facilities security product, contributing Selenium frontend automated tests and UI/UX improvements.
- Tested software interactions with security scanners by configuring and running automated tests in Linux terminals.
- Ensured product specifications met security standards and maintained quality by developing detailed test plans in an Agile environment.

PERSONAL PROJECTS

Personal Website Platform

January 2026 - Present

- Designed and deployed a full-stack personal web platform using React, TypeScript, Node.js, and Azure cloud services.
- Implemented backend service hosting, environment configuration, API deployment, and PostgreSQL integration on Azure.
- Established CI/CD workflows for automated deployment and infrastructure management.
- Configured application monitoring, logging, and operational tooling to support ongoing maintenance and troubleshooting.
- Managed the complete deployment lifecycle independently, including cloud provisioning, configuration, deployment, and production support.

SKILLS

Languages: TypeScript/JavaScript, C++, C#, Java, Python

Frontend: React, Angular, TailwindCSS, Zod, Zustand

Backend and Architecture: Node.js API Development, .NET 8+, Cloud Deployment, Telemetry & Monitoring

Cloud and Infra: Git, Azure App Services, Azure PostgreSQL, Azure Application Insights, CI/CD Pipelines, Environment Configuration Management, REST API Deployment, SQL

Testing and Automation: Playwright, Selenium, React Testing Library, Jest, Postman, Swagger

AI-Assisted Development: GitHub Copilot, Large Language Models, Generative AI, OpenAI API, Retrieval-Augmented Generation, Vector Databases

EDUCATION

Rochester Institute of Technology - BS in Game Design and Development

August 2015 - May 2019